DIAGNOSCAN – INTELLIGENT DIAGNOSIS THROUGH MACHINE LEARNING

Balancing Innovation and Responsibility

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PROBLEM STATEMENT

What's the challenge?

Diagnosing medical conditions is complex and time-consuming

Risk of human error in early-stage diagnostics

Need for a tool that supports decision-making with data-driven insights

OUR SOLUTION – DIAGNOSCAN

A machine learning-powered diagnostic assistant

Accepts structured patient reports

Predicts probable diagnosis using Support Vector Machine (SVM)

Presents interpretable visualizations to enhance trust and clarity

HOW IT WORKS

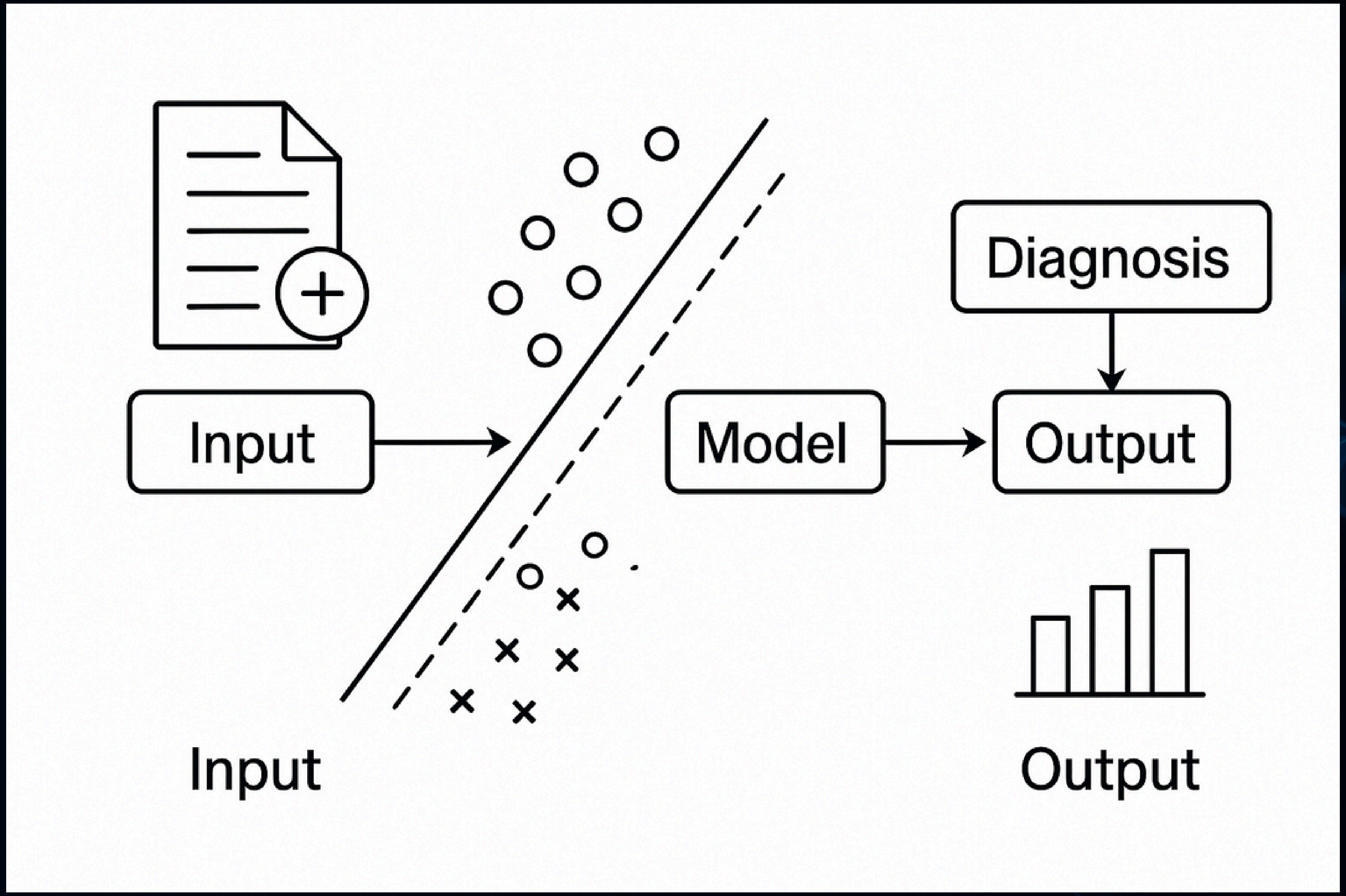
Input: Patient report with relevant metrics (e.g., blood values, symptoms)

Model: Trained SVM classifier analyzes patterns in patient data

Output:
Likely diagnosis.
Confidence score / probability.
Comparative metrics.
Visual insights.

PROCESS


Compare
Indicator
Template
Snap
Save
Start
5%
Analysis
30%
Plan
18%
Idea
85%
Brainstorm
50%


Diagnosis
Input
Model
Output
Input
Output

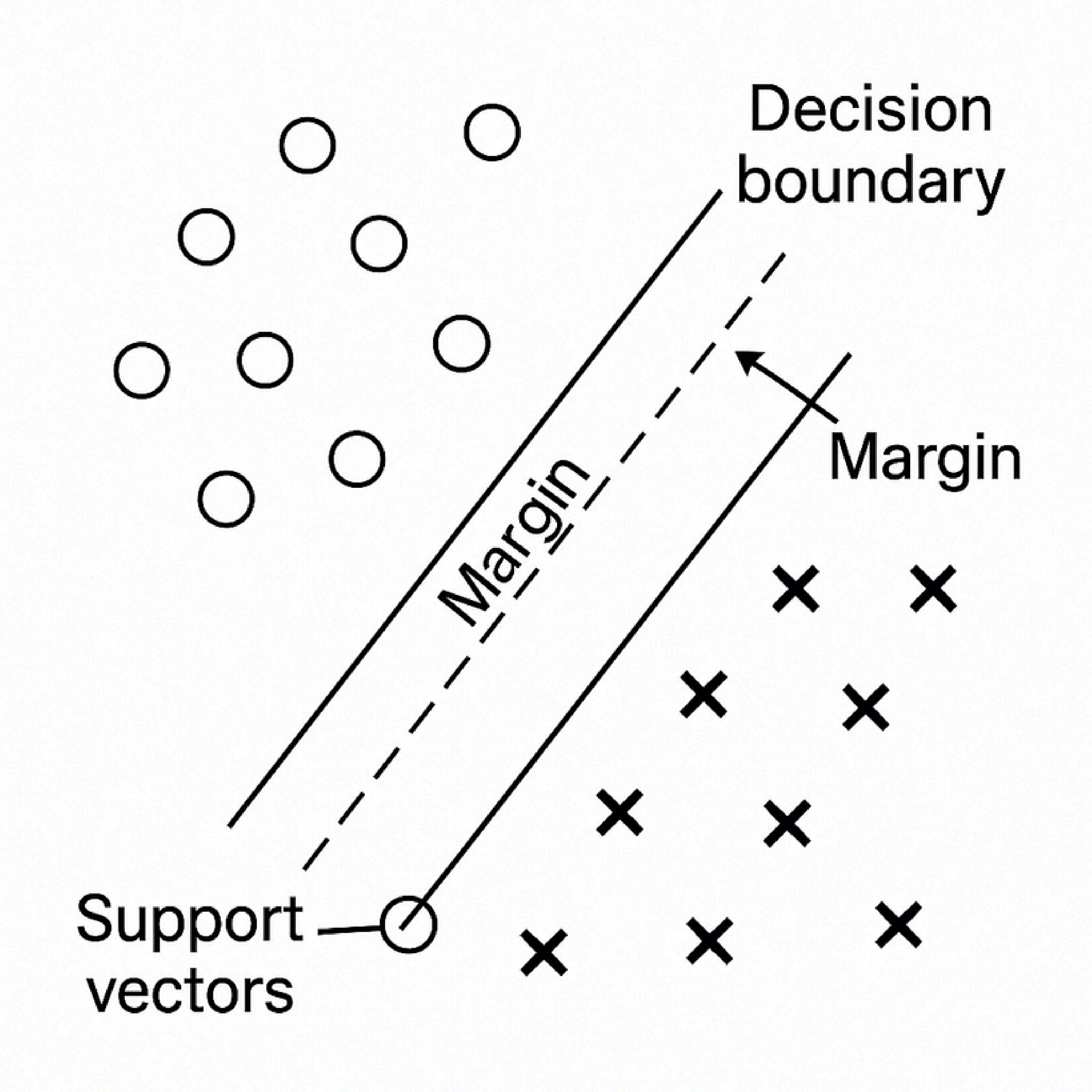
WHY SVM?

Effective for high-dimensional data

Strong in binary classification (healthy vs. condition)

Robust to outliers

Performs well with limited samples, common in medical datasets



IMPACT AND BENEFITS

- Faster pre-screening of cases
- Better support for physicians in decision-making
- Transparency through explainable visualizations
- Scalable across different conditions and report types

CASE STUDY / DEMO

FUTURE ROADMAP

Expand to multi-condition diagnoses.

Integrate with hospital EHRs

Use ensemble models for better accuracy

Enable multilingual and voice-based inputs

THANK YOU!

FOR YOUR ATTENTION